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**Course Name:**5G 6G Communication Systems

**Course Code:**ECA5601

### **Ex:13: V2X Latency Comparison**

#### **Aim:**

To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

#### **Objective:**

- 1.To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.
- 2.To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.
- 3.To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

#### **Procedure:**

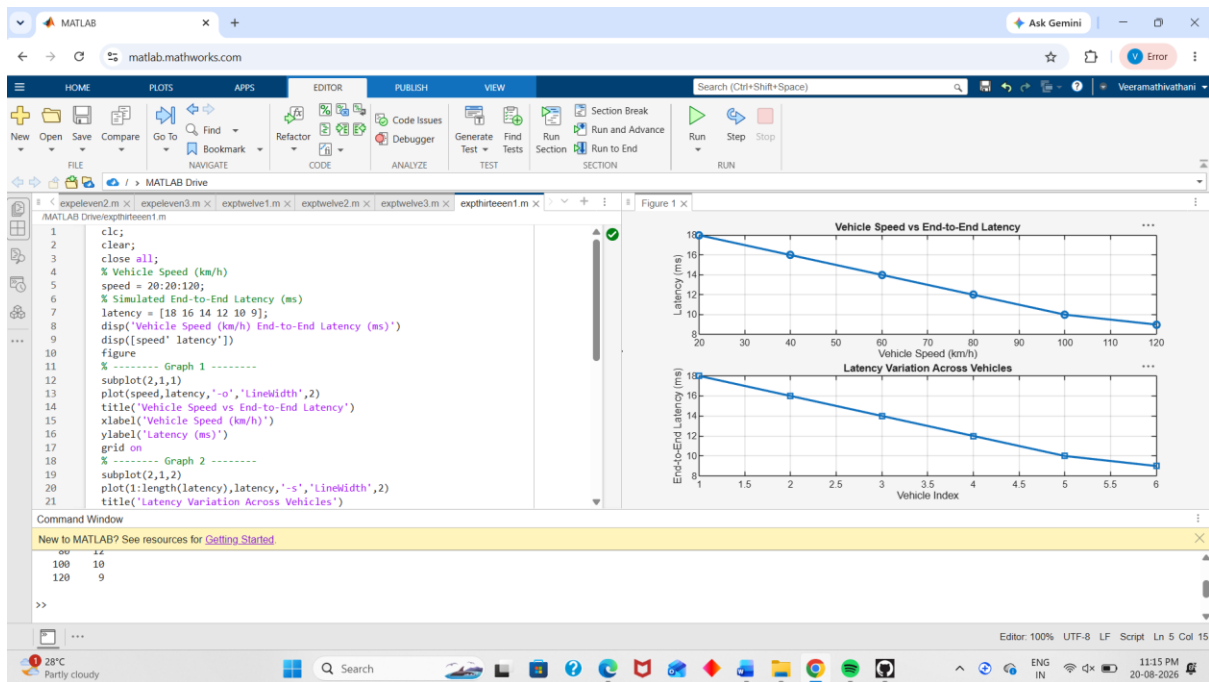
1. Define the V2X communication parameters such as vehicle speed, transmission range, packet size, and simulation time.
- 2.Generate different vehicle movement scenarios by varying the vehicle speed.
- 3.Simulate wireless communication between vehicles using a V2X communication model.
- 4.Calculate the End-to-End Latency (ms) for data transmission between communicating vehicles.
- 5.Compute the Packet Delivery Ratio (PDR %) by comparing the number of successfully received packets with transmitted packets.
- 6.Analyze the effect of increasing Vehicle Speed (km/h) on communication latency and packet delivery performance.

7. Plot MATLAB graphs for End-to-End Latency, Packet Delivery Ratio, and Vehicle Speed.

8. Compare the obtained results to evaluate the efficiency and reliability of the V2X communication system under different traffic conditions.

### **Objective 1: Matlab Code and Output**

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```



## Objective 2: Matlab Code and Output

`clc;`

`clear;`

`close all;`

`% Vehicle Speed (km/h)`

`speed = 20:20:120;`

`% Simulated Packet Delivery Ratio (%)`

`PDR = [99 98 97 95 93 90];`

`disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')`

`disp([speed' PDR'])`

`figure`

`% ----- Graph 1 -----`

`subplot(2,1,1)`

`plot(speed,PDR,'-o','LineWidth',2)`

`title('Vehicle Speed vs Packet Delivery Ratio')`

`xlabel('Vehicle Speed (km/h)')`

`ylabel('Packet Delivery Ratio (%)')`

grid on

% ----- Graph 2 -----

subplot(2,1,2)

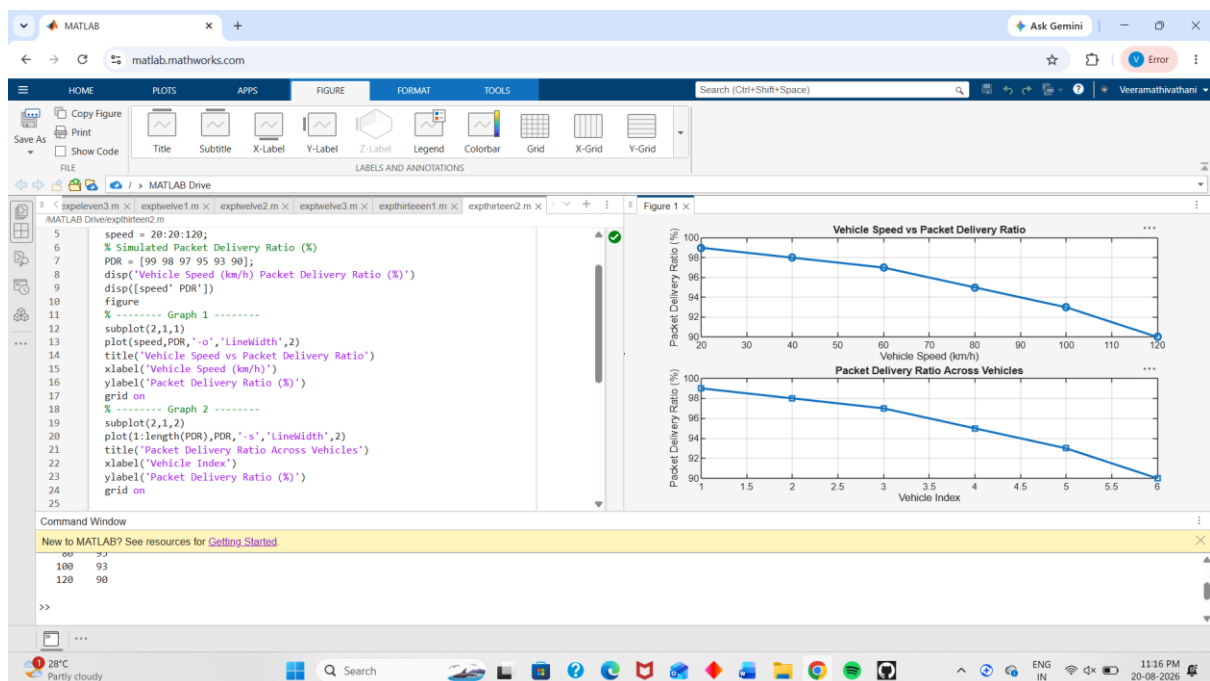
plot(1:length(PDR),PDR,'-s','LineWidth',2)

title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')

grid on



### Objective 3: Matlab Code and Output

clc;

clear;

close all;

% Vehicle IDs

vehicle = 1:6;

% Vehicle Speed (km/h)

speed = [30 45 60 75 90 105];

% Communication Range (m)

range = [280 260 240 220 200 180];

```
disp('Vehicle Speed(km/h) Communication Range(m)')
```

```
disp([vehicle' speed' range'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(vehicle,speed,'-o','LineWidth',2)
```

```
title('Vehicle Speed of Different Vehicles')
```

```
xlabel('Vehicle Index')
```

```
ylabel('Vehicle Speed (km/h)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

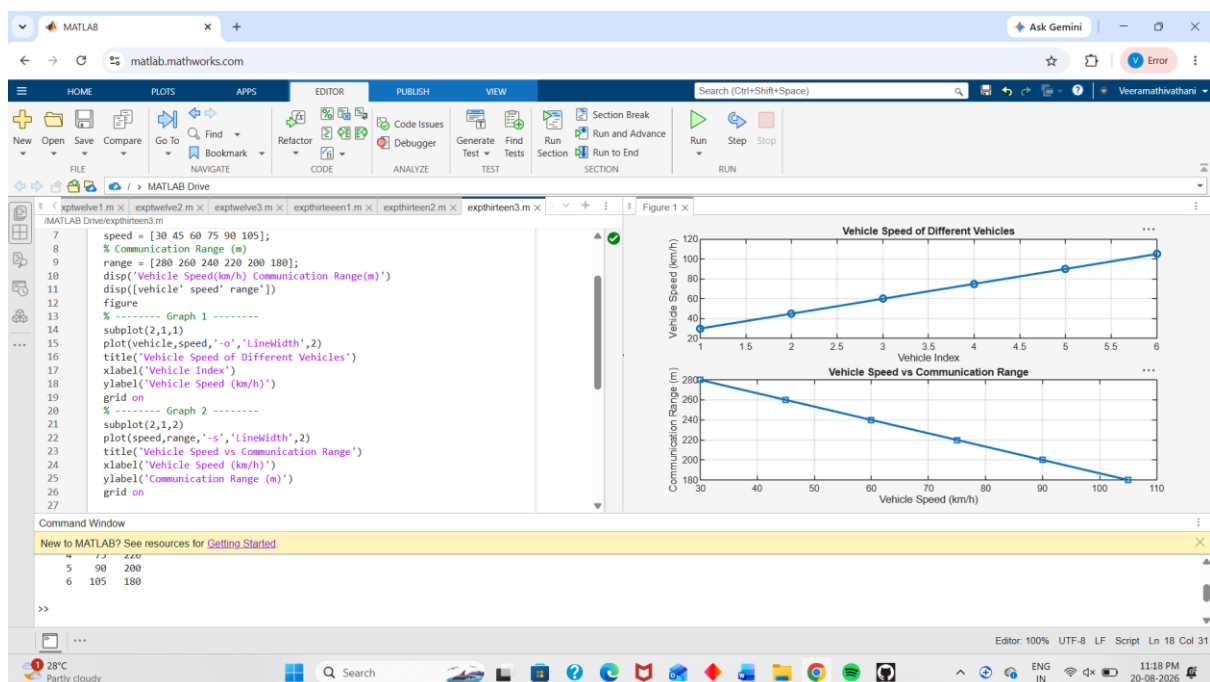
```
plot(speed,range,'-s','LineWidth',2)
```

```
title('Vehicle Speed vs Communication Range')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Communication Range (m)')
```

```
grid on
```



**Result:**

1. The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.
2. The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.
3. The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.